

M19 Teaching Book

Self-contained teacher-style interview/job-launch instruction

Standalone Zero-to-Job-Ready Data Engineer 2026 | M19

Why M19 is practice-led

Interview competence is retrieval and transfer. For this module, a tutorial immediately before the attempt would often inflate recognition. The Teaching Book teaches the framework first; the actual timed attempt is closed-book; Review opens only after saved evidence.

DE19-01 - Truthful one-page Data Engineer resume

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

Translate course evidence into truthful claims. A resume bullet is valid only when you can open the repo/output that proves it.

WORKING MODEL

Evidence -> action -> system -> validation/result. Example: Built a Python ingestion pipeline that quarantines invalid rows and proved idempotent reruns with unchanged trusted counts.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Inventing employment, years of experience, production scale, users, revenue impact, certifications or cloud runtime you did not actually have.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-01.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-01 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-01.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-01.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain truthful one-page data engineer resume in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-01_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-02 - GitHub and project portfolio final polish

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

A portfolio is a recruiter navigation problem and a reproducibility problem, not a decoration contest.

WORKING MODEL

Pinned repo -> README -> architecture -> setup -> test/evidence -> failure/recovery -> limitations.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Polished screenshots with broken setup, copied tutorial repos, hidden secrets, dead files, or claims that exceed executed evidence.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-02.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-02 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-02.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-02.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain github and project portfolio final polish in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-02_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-03 - LinkedIn/professional profile without overclaiming

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

Your public profile should tell a coherent transition story without converting learning into fake employment.

WORKING MODEL

Target role + defensible stack + strongest proof + current availability. Keep claims narrow enough to defend in a live interview.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Using “Data Engineer at X” when you are not employed there, invented years, or stuffing every technology name into the headline.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-03.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-03 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-03.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-03.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain linkedin/professional profile without overclaiming in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-03_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-04 - Degree-aware job targeting: apply/skip rules

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

Job targeting is an eligibility and expected-skill matching exercise. Read required vs preferred literally.

WORKING MODEL

APPLY when core work matches and hard requirements are met; MAYBE when a gap is preferred/trainable; SKIP when a hard legal/degree/experience requirement clearly excludes you.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Mass-applying to roles requiring unrelated seniority or mandatory qualifications, then misreading rejections as proof the field is closed.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-04.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-04 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-04.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-04.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain degree-aware job targeting: apply/skip rules in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-04_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-05 - SQL interview drills

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

SQL interviews test translation from business question to grain, joins, filters, windows and validation under time pressure.

WORKING MODEL

Question -> state grain -> write query -> inspect duplicates/nulls -> validate with an independent count/total -> explain trade-off.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

A query that runs but has wrong join cardinality, denominator, filter timing or nondeterministic ranking.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-05.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-05 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-05.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-05.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain sql interview drills in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-05_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-06 - Python interview drills

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

Python interviews for junior DE roles usually reward readable data manipulation, validation, error handling and testable behavior more than clever tricks.

WORKING MODEL

Input contract -> transform -> explicit failure path -> deterministic output -> edge-case tests -> complexity at a practical level.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Silent dropping, mutable hidden state, catching every exception, or code that cannot be explained line by line.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-06.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-06 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-06.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-06.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain python interview drills in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-06_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-07 - Data-engineering concepts and troubleshooting drills

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

Troubleshooting answers are strongest when they start from evidence and preserve safe state.

WORKING MODEL

Symptom -> blast radius -> first evidence -> likely layer -> smallest safe test -> repair -> rerun boundary -> reconciliation -> prevention.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Restarting/deleting/reprocessing everything before understanding persisted state, or claiming a cause without logs/metrics/data evidence.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-07.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-07 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-07.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-07.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain data-engineering concepts and troubleshooting drills in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-07_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-08 - Project walkthrough interview

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

A project walkthrough proves ownership. Interviewers will probe grain, failure handling, validation and why each tool exists.

WORKING MODEL

Problem -> sources -> architecture -> grain -> quality -> idempotency -> orchestration -> failure/recovery -> security -> limitations -> 10x change.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Reciting README buzzwords, reading code to remember basic design, or saying “production-ready” without production evidence.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-08.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-08 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-08.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-08.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain project walkthrough interview in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-08_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-09 - Architecture/system-design defense for junior roles

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

Junior system design is about requirements and trade-offs, not drawing the biggest possible architecture.

WORKING MODEL

Requirements -> scale/latency -> source contracts -> storage/compute -> orchestration -> correctness -> security -> observability -> failure modes -> cost/operability.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Choosing Kafka/Spark/Kubernetes by default when batch SQL/Python would meet the requirement more safely.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-09.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-09 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-09.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-09.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain architecture/system-design defense for junior roles in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-09_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-10 - Behavioral answers and career-transition story

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

Behavioral answers should be truthful evidence stories, including failures and corrections.

WORKING MODEL

Context -> responsibility -> action -> result -> verification -> reflection. Use course/project facts if you do not yet have professional DE experience.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Invented workplace stories, blaming teammates/tools, or giving a perfect story with no decision, result or lesson.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-10.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-10 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-10.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-10.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain behavioral answers and career-transition story in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-10_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-11 - Technical English: explain code, failures and decisions clearly

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

Technical English is clarity of reasoning, not accent. Simple nouns and verbs beat memorized jargon.

WORKING MODEL

State what the object is, what changed, why, how you checked it, and one way it can fail.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Long definitions with no concrete example, rushing when stuck, or hiding uncertainty instead of correcting yourself calmly.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-11.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-11 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-11.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-11.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain technical english: explain code, failures and decisions clearly in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-11_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-12 - Application tracker, feedback loop and weekly repair cycle

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

A job search is a measurable funnel. Change the likely bottleneck rather than changing everything after each rejection.

WORKING MODEL

Targeted role -> truthful application -> screen -> assessment -> technical -> final -> offer. Record stage-specific failure patterns and run one repair experiment per batch.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

No tracker, repeated chasing, mass ineligible applications, or endless course polishing instead of testing the market.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-12.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-12 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-12.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-12.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain application tracker, feedback loop and weekly repair cycle in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-12_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-13 - Timed coding test: arrays, strings and hash-map problems

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

Coding screens protect basic problem decomposition and data-structure fluency; this is not a competitive-programming syllabus.

WORKING MODEL

Restate contract -> examples -> choose data structure -> implement -> edge cases -> complexity -> verbal explanation.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Memorizing solutions, ignoring empty/duplicate cases, or using an $O(n^2)$ pattern without noticing when a hash map gives a clearer $O(n)$ approach.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-13.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-13 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-13.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-13.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain timed coding test: arrays, strings and hash-map problems in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-13_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.

DE19-14 - Combined SQL + Python interview simulation

OUTCOME

Produce a defensible artifact or timed answer that proves this competency under realistic junior Data Engineer conditions.

TEACH

The final mixed mock tests context switching: SQL, Python, troubleshooting and project defense under a bounded clock.

WORKING MODEL

Save first attempt -> classify misses -> open review -> repair exact competency -> close review -> fresh changed mixed mock -> defend without notes.

WHY IT MATTERS

Employers cannot observe what you studied. They observe whether you can produce, debug, explain and defend evidence without inventing experience.

COMMON FAILURE

Repeating the same mock after reading answers and calling recognition “mastery”.

GUIDED REPRODUCTION

Open 02_GUIDED/GUIDED_DE19-14.md. Follow the scaffold once using the guided scenario. Save the output, then close it.

DE19-14 - Transfer and mastery

INDEPENDENT ATTEMPT

Open 03_INDEPENDENT/INDEPENDENT_DE19-14.md. Do the changed task without the guided answer open. Save first evidence before Review.

VALIDATE

Check factual truthfulness, technical correctness, evidence traceability and whether you can explain the result without reading the artifact.

RECOVER

If weak, classify the miss: knowledge, reasoning, syntax, validation, communication, time pressure or claim/evidence mismatch. Repair only that dimension.

FRESH RETRY

After Review, close it and complete 06_FRESH_RETEST/RETEST_DE19-14.md. The retry changes the domain/data/prompt.

EXPLAIN BACK

Explain combined sql + python interview simulation in 60-120 seconds with one concrete example, one failure risk and one verification step.

FINISHED WHEN

Save 07_EVIDENCE/DE19-14_EVIDENCE.md with first-attempt path, validation, repair if any, fresh-retry path and explain-back note.